Code-Layout, Readability and Reusability

Date	15 March 2024
Team ID	unm293200909
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Name	Kavyasri.A
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the FitBuddy codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	4-space Python indentation in all .py files; consistent HTML indentation in templates.
2	Proper File Structure	Files and folders are logically organized	Yes	/app for Python files, /templates for HTML, /static/images for assets.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names such as goal, intensity, original_plan, updated_plan describe their data.
4	Function / Method Names	Functions are descriptively named	Yes	generate_workout_gemini(), generate_nutrition_tip_with_flash(), update_workout_plan(), save_user().
5	Code Comments	Inline and block comments explain logic	Yes	Comments explain prompts, routes and database operations.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules: gemini_generator.py, gemini_flash_generator.py, updated_plan.py, database.py, routes.py.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Each AI task and database action has one function reused by the routes.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic handling of failed Gemini calls and missing users; can be extended.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	generate_workout_gemini()	Python, Gemini 1.5 Pro API	Plan creation route; result.html display	High
2	generate_nutrition_tip_with_flash()	Python, Gemini Flash API	Tip shown with the original and the updated plan	High

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
3	update_workout_plan()	Python, Gemini 1.5 Pro API	Feedback route to regenerate the plan	Medium
4	database.py (save_user, save_plan, get_user, ...)	Python, SQLAlchemy, SQLite	All routes and the admin view of users	High
5	Jinja2 templates (index, result, all_users)	HTML, CSS, Jinja2	Rendered by every route through TemplateResponse	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	4	Clear folders and separate modules for AI, database and routes.
Readability	4	Descriptive names and simple, consistent formatting.
Reusability	4	AI and database functions are reused across routes and templates.
Documentation / Comments	3	Comments present; docstrings and a README can be added.
Overall Score	4	Average of the four ratings is 3.75 out of 5.

Work Assignment:

Work Assigned	Assigned To
Review indentation, naming and file structure	Kavyasri.A
Add comments / docstrings and improve error handling	Kavyasri.A
Identify and document reusable modules	Kavyasri.A
Rate code quality and prepare this report	Kavyasri.A